

Resumes: Courage — Answers

Courage

The abstract data generating mechanism coming out of Justice is:

$$\text{logit}(\Pr(Y = 1)) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$$

Candidate Models

Estimate a model in which `call` is the outcome and `city` is the only covariate. Display the coefficients and their confidence intervals.

```
logistic_reg() |>
  fit(call ~ city, data = resumes) |>
  tidy(conf.int = TRUE)
```

```
# A tibble: 2 x 7
  term          estimate std.error statistic    p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)   -2.23      0.0726    -30.7 2.22e-207 -2.38    -2.09
2 citychicago -0.397     0.106     -3.76 1.70e- 4 -0.605   -0.190
```

The coefficient on `citychicago` is about -0.40 with a 95% confidence interval excluding zero: callbacks are less likely in Chicago than in Boston. Location matters, but it says nothing yet about the question both scenarios actually care about — so we add the name's ethnicity next.

Estimate a model in which `call` is the outcome and `city` and `ethnicity` are the covariates. Display the coefficients and their confidence intervals.

```
logistic_reg() |>
  fit(call ~ city + ethnicity, data = resumes) |>
  tidy(conf.int = TRUE)
```

A tibble: 3 x 7

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>	conf.low <dbl>	conf.high <dbl>
1	(Intercept)	-2.47	0.0964	-25.6	5.88e-145	-2.66	-2.29
2	citychicago	-0.399	0.106	-3.77	1.66e- 4	-0.607	-0.191
3	ethnicitycauc	0.439	0.107	4.09	4.33e- 5	0.230	0.652

The coefficient on `ethnicitycauc` is about 0.44 with a confidence interval excluding zero — a white-sounding name raises the log-odds of a callback, and the city coefficient barely moves, which is what randomized name assignment should produce. The paper’s second headline finding is that resume quality pays off more for white-sounding names, so we test that next.

Add `quality`, and its interaction with `ethnicity`, to the previous model. Display the coefficients and their confidence intervals.

```
logistic_reg() |>
  fit(call ~ city + ethnicity + quality + ethnicity * quality, data = resumes) |>
  tidy(conf.int = TRUE)
```

A tibble: 5 x 7

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>	conf.low <dbl>	conf.high <dbl>
1	(Intercept)	-2.43	0.125	-19.4	4.00e-84	-2.68	-
2	citychicago	-0.400	0.106	-3.78	1.59e- 4	-0.608	-
3	ethnicitycauc	0.523	0.147	3.55	3.81e- 4	0.237	0.814
4	qualitylow	-0.0879	0.165	-0.531	5.95e- 1	-0.413	0.236
5	ethnicitycauc:qualit~	-0.179	0.216	-0.832	4.05e- 1	-0.603	0.244

Both the `quality` main effect and its interaction with `ethnicity` have confidence intervals that comfortably include zero. The paper’s pattern does not show up in this simple model, so we drop `quality` and instead keep covariates whose intervals exclude zero — plus `sex`, which the historian’s question forces us to include.

The Data Generating Mechanism

Which model should we choose, and why? Define `fit_resumes` as your chosen model and write out the fitted equation.

```
fit_resumes <- logistic_reg() |>
  fit(call ~ sex + city + ethnicity + special, data = resumes)

fit_resumes |>
  tidy(conf.int = TRUE)
```

```
# A tibble: 5 x 7
  term          estimate std.error statistic    p.value conf.low conf.high
<chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)   -2.75      0.118    -23.3  1.56e-120 -2.98    -2.52
2 sexmale       -0.126     0.133    -0.950  3.42e- 1 -0.392    0.130
3 citychicago  -0.463     0.108    -4.28   1.85e- 5 -0.676   -0.252
4 ethnicitycauc  0.447     0.108     4.13   3.64e- 5  0.236    0.660
5 specialyes     0.817     0.108     7.57   3.72e-14  0.606    1.03
```

We choose `call ~ sex + city + ethnicity + special`. The `city`, `ethnicity`, and `special` (special skills) coefficients all have confidence intervals excluding zero. The `sex` coefficient's interval includes zero, but the historian's question explicitly asks about differences by sex, so the question — not the data — forces it into the model. Quality stays out, per the previous model.

$$\text{logit}(\Pr(\text{call} = \text{yes})) = -2.75 - 0.13 \cdot \text{male} - 0.46 \cdot \text{chicago} + 0.45 \cdot \text{caucasian} + 0.82 \cdot \text{special}$$

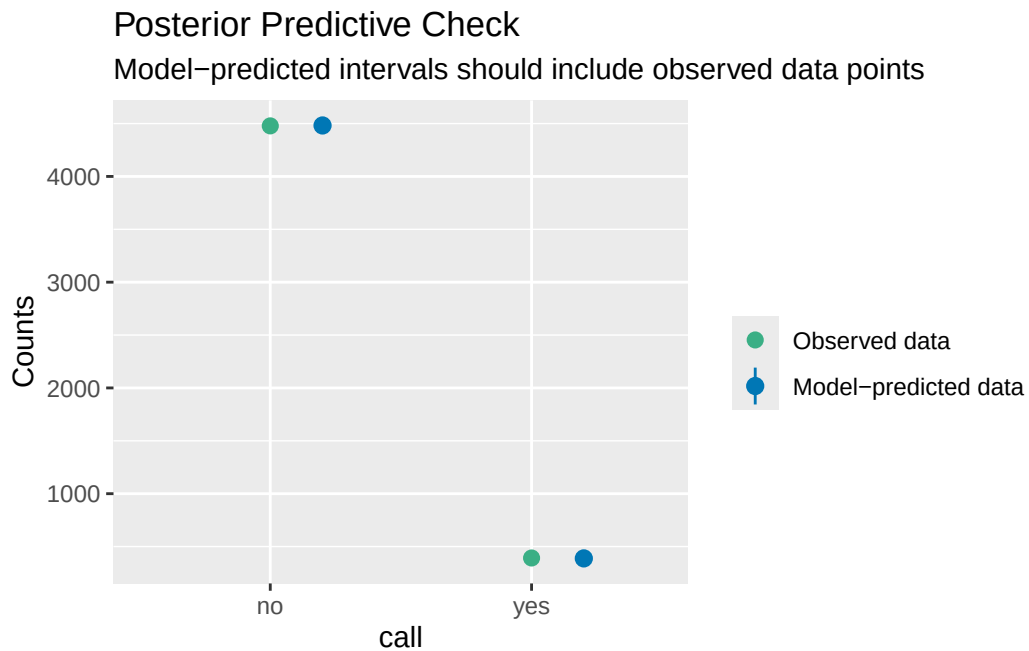
Each indicator is 0/1. The intercept describes a female applicant in Boston with a black-sounding name and no special skills; each coefficient shifts the log-odds of a callback. **This is our data generating mechanism.**

Model Checking

Use `check_predictions(extract_fit_engine(fit_resumes))` to perform a posterior predictive check of your chosen model.

```
check_predictions(extract_fit_engine(fit_resumes))
```

```
Ignoring unknown labels:  
* size : ""  
* alpha : ""
```



`check_predictions()` simulates new callback data from the fitted model and compares it to the data we observed. The simulated replicates bracket the observed share of callbacks, so the model is not obviously misdescribing the outcome — the check we want to pass before trusting the predictions in Temperance.